Predicting myopia in medical students

Choosing models with 132 students: what worked, and how much of it to trust

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One dataset, four questions about myopia

M

Does the student have myopia?

SI / NO

MM

Myopia magna, over 6 dioptres?

SI / NO

Combo

Control, myopia or myopia magna?

C / M / MM

DCombo

Same, with myopia split in two

C / M1 / M2 / MM

Goal: labels for 24 test students with no answers, as accurate as possible, with a process we can defend.

Few students, few cases, and messy columns

132

training students, and 24 for the test

10

with MM = SI: always saying NO scores 0.92

49 to 33

raw columns to numeric features, each one justified

Phototype: a mixed-up column

Letters from a neighbouring column were typed in it, so 45 % of the training values are missing once they are removed.

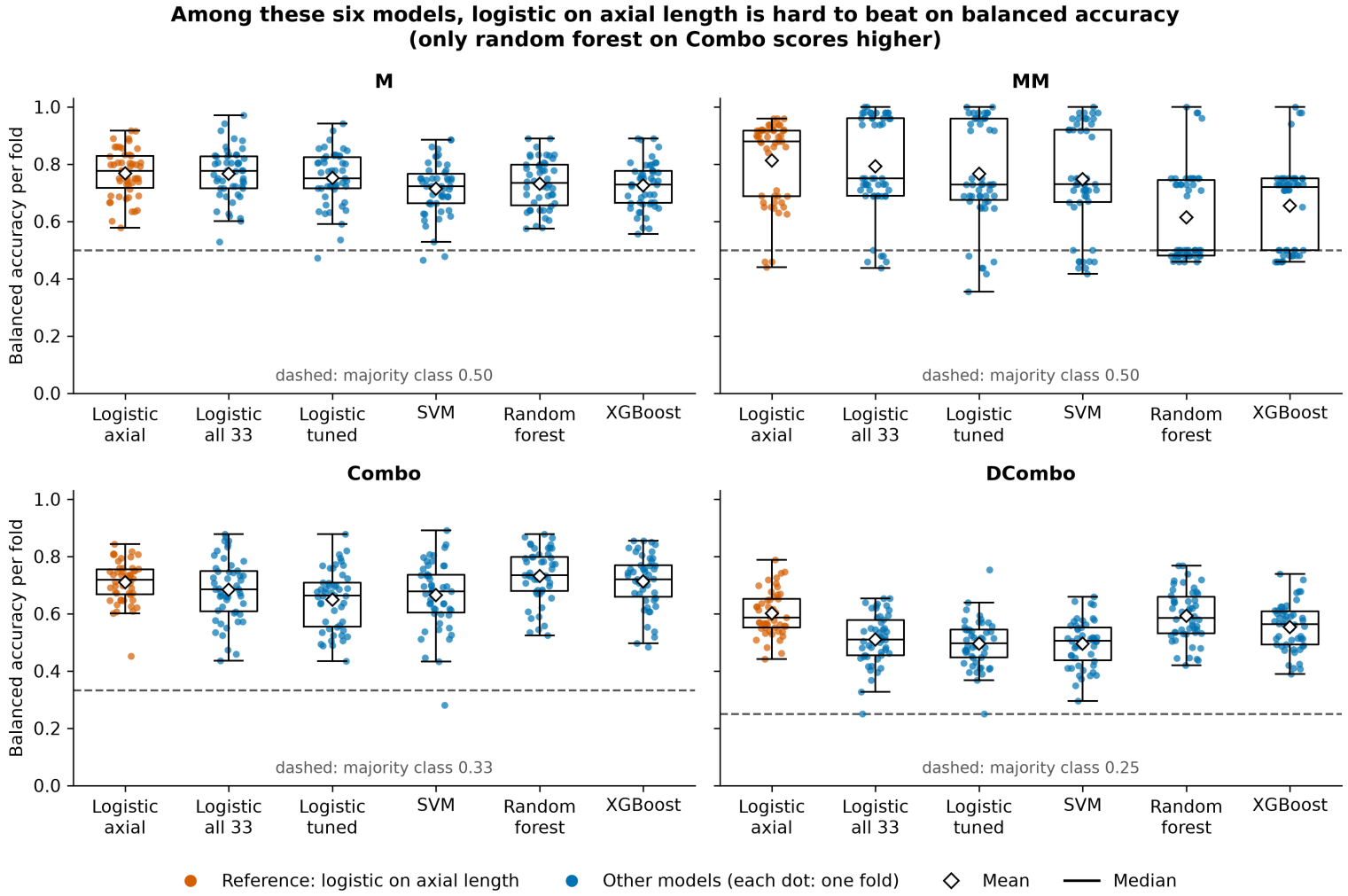
Eight students who contradict themselves

They have Combo = MM but MM = NO. We tried correcting them, and kept the labels as provided: only the drop in accuracy was firm.

With so little data, the method matters most

1. **Repeated CV** — 5 folds, 10 repeats. MM has exactly 2 positives per fold, so one split would mean little.
2. **No leakage** — Medians, scaling and tuning are learned inside the training folds, with nested CV.
3. **Written test** — A model must clearly beat the baseline on balanced accuracy without being clearly worse on macro-F1, or pass a second route.
4. **One test use** — The 24 test students are used once, to predict, and nothing is chosen from them.

Complex models rarely beat a one-feature baseline



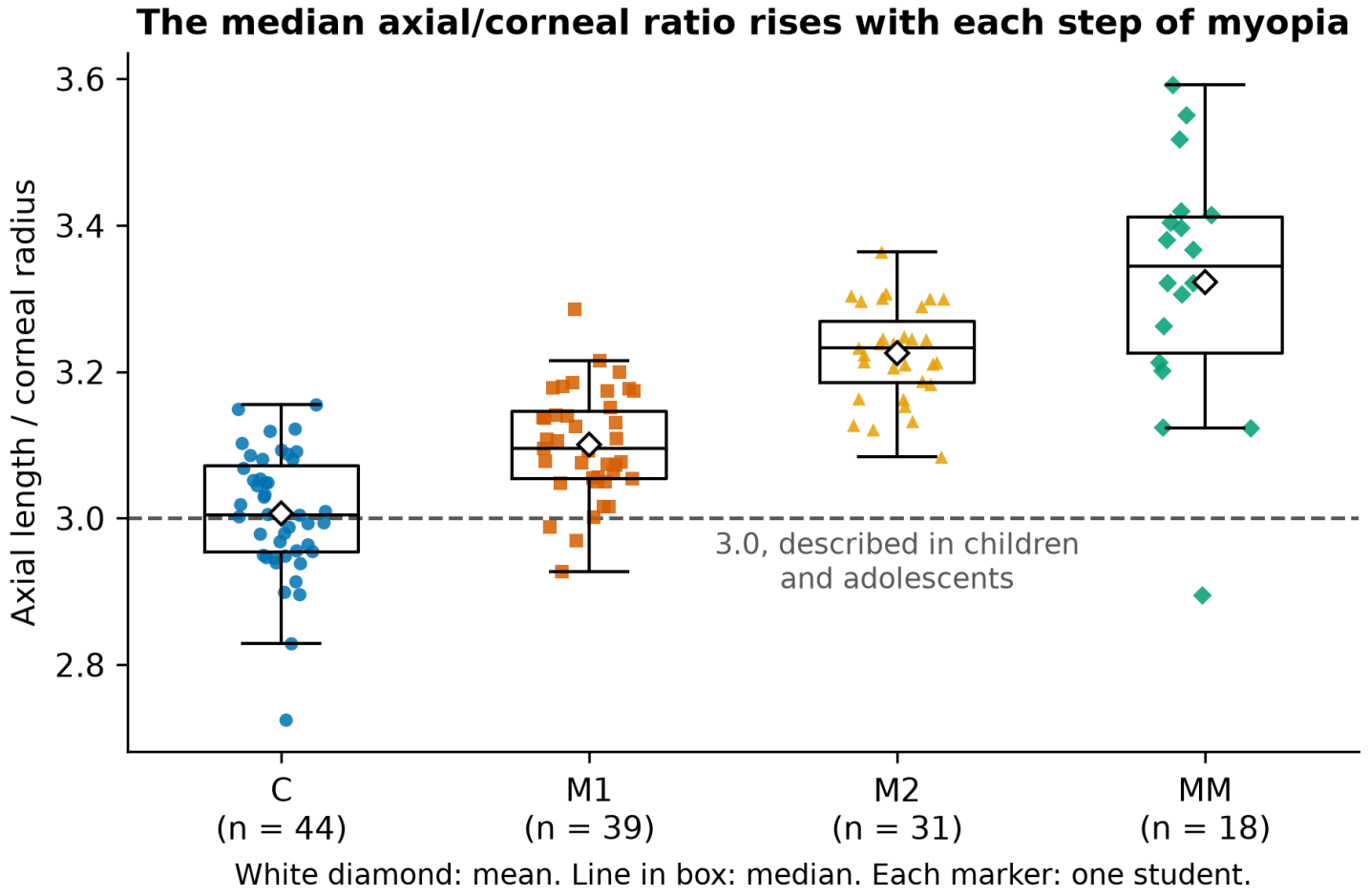
XGBoost (new to us)

Builds small trees one after another, each one correcting the errors of the previous ones.

We limited it to shallow trees and a small grid, to avoid overfitting.

It never came first.

Medicine gave us what complexity did not



Axial length / corneal radius

The eye focuses through its length and its cornea together.

Computed for each student, without labels.

Units not stated in the brief: we deduced dioptres from the data.

Yebra-Pimentel et al., Arch Soc Esp Oftalmol 2004; Fan, Huang and Huang, J Ophthalmol 2024.

With the ratio, a new model won on every target

Target	Final model	Balanced acc.	Macro-F1	Accuracy
M	Logistic: axial + ratio	0.77 → 0.80	0.75 → 0.78	0.77 → 0.80
MM	Logistic: ratio, tuned threshold	0.81 → 0.89	0.65 → 0.88	0.82 → 0.97
Combo	Random forest + ratio	0.71 → 0.77	0.66 → 0.77	0.65 → 0.75
DCombo	Logistic: axial + ratio	0.60 → 0.65	0.58 → 0.64	0.56 → 0.63

From logistic regression on axial length alone to the final model. Validation scores, before correcting for selection.

Tried and not chosen: an ordinal model for DCombo. It passed the test, but scored below logistic regression with the ratio.

On MM, accuracy has a price: missed cases

MM model on the ratio	Balanced accuracy	Macro-F1	Accuracy	Recall of SI
Threshold 0.5	0.89	0.75	0.88	0.90
Tuned threshold (chosen)	0.89	0.88	0.97	0.79
Always saying NO	0.50	0.48	0.92	0.00

For this assignment

The tuned threshold: it passed the second route of our test, and the test set is graded on accuracy.

For a clinic

A threshold of 0.5 finds about one more of the 10 positives, with about 12 more false alarms. The choice depends on the cost of each error.

The cornea may explain the eight odd students

Group	Students	Axial length (mm)	Keratometry (D)	Ratio
MM = SI	10	26.35	43.40	3.39
The 8 inconsistent	8	26.52	41.28	3.24
Combo = M	70	24.47	43.55	3.16
Combo = C	44	23.47	43.27	3.01

Same eye length as MM = SI, but flatter corneas. The cornea may partly compensate, keeping them under 6 dioptres.

A hypothesis: there are no refraction values in the data, and their ratios overlap with the MM = SI group.

We measured how optimistic our scores are

Balanced accuracy	Notebook score	Repeating all the selection	Axial length alone
M	0.80	0.79	0.77
MM	0.89	0.82	0.81
DCombo	0.65	0.59	0.59

Combo is left out: this estimate uses a simpler forest than the chosen one. It is an approximation, with fewer students in training.

The cornea seems to carry real information. The exact gains, and the choice of model on Combo, are uncertain.

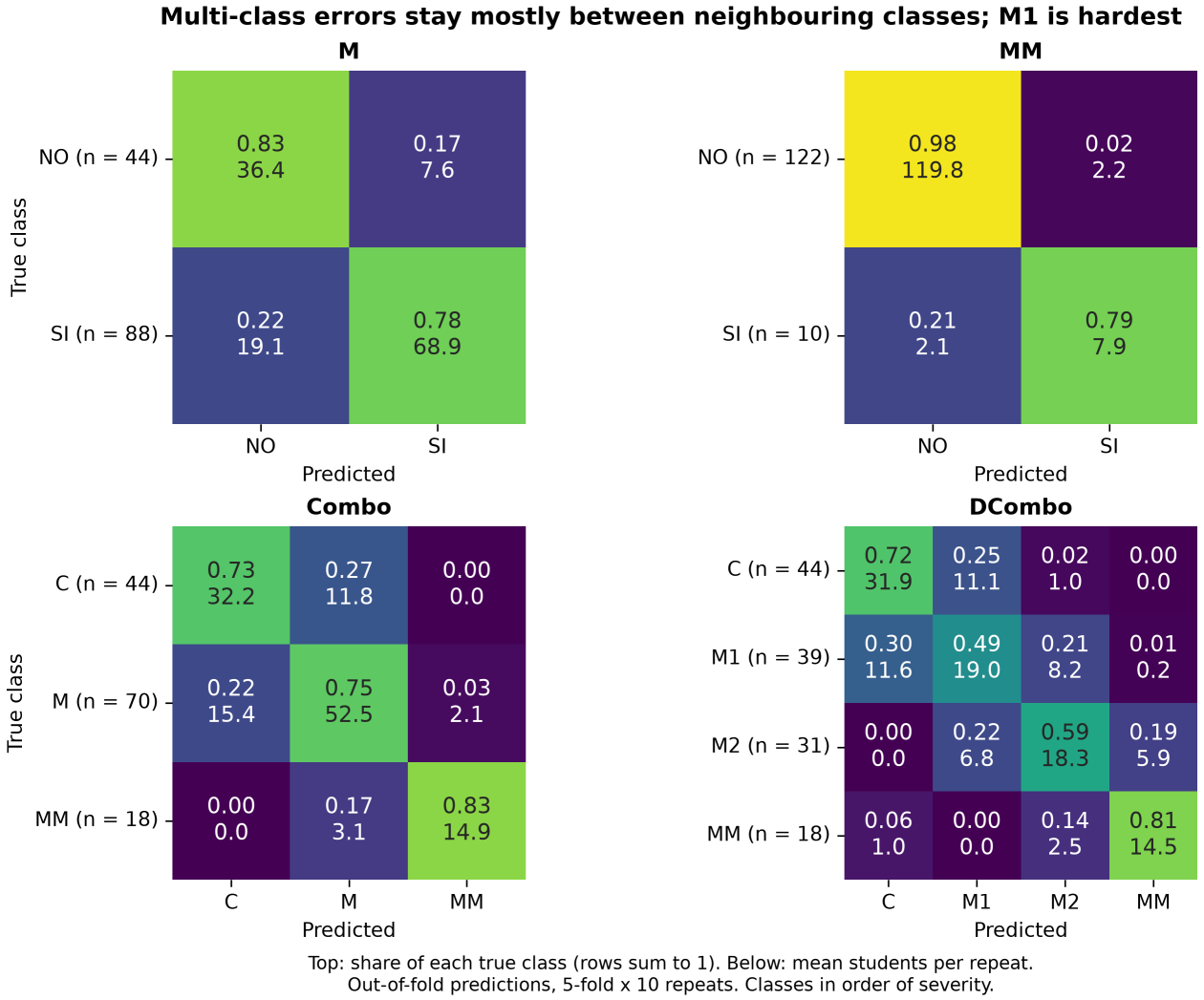
1 in 3

A useless feature passes our test on Combo about this often.

42

selection tests on the same 132 students, all reported.

Most errors fall between neighbouring classes



Top: share of each true class (rows sum to 1). Below: mean students per repeat.
Out-of-fold predictions, 5-fold x 10 repeats. Classes in order of severity.

63 / 35 / 2

DCombo, %: correct, off by one class, off by more

What the models use

Almost only axial length and the ratio, even in the forest. That describes the models, not the causes of myopia.

What we expect on the test set

Accuracy	Majority class	Sober estimate	Optimistic estimate
M	0.67	0.79	0.80
MM	0.92	0.92	0.97
Combo	0.53	0.68	0.75
DCombo	0.33	0.57	0.63

One student is worth about 0.04. Sampling alone moves a score by about 4 to 5 students, or 2 to 3 on MM. The Combo sober figure uses a simpler forest.

5 of 24

Test students predicted MM = SI, against 8 % in training.

We did not change them: that would be choosing with the test set.

What we take away

Domain knowledge helped more than complexity

In our comparisons, a ratio from the literature did what added model complexity did not.

Measuring doubt is part of the result

A nested estimate and a useless-feature check show how far to trust the scores.

The limit is partly the data

More students and a direct measure of refraction would help more than another model.

Team and tools

Who did what

All parts were done together, taking the decisions jointly: Ana Gomez and Tomás Oriol Freixa.

Workload

50/50

Split evenly between the two members.

Keras / PyTorch

Neither

We used scikit-learn and XGBoost. With 132 students, neural networks were hard to justify.

How we used generative AI

What the AI did (Claude Code)

Generated much of the code, and drafted the notebook text.

Answered doubts and conceptual questions.

Proposed approaches, candidate models and selection rules, and found the literature on the ratio.

Reviewed methodology, charts and consistency with review agents.

Built these slides and drafted the speaker notes.

What we did

Took the decisions, such as keeping the labels and accepting the rules.

Proposed ideas to the AI about what could be done.

Asked for the reviews and decided what to change.

Kept a log of the AI use, in docs/ai_usage.md.